GinCol Lab · Aula

Student worksheet

Mathematics I · 1st year of Bachillerato · The real numbers · Class 5 of 6 · With Ximena

Logarithms: the power in reverse

Class video: https://youtu.be/QalMQUfeaBw

The code leads to the video.

Name: _____ Date: _____

The video is in Spanish, with English subtitles. In Spain the decimal separator is a comma: the board writes 0,01 for 0.01. This sheet uses the decimal point.

Before watching the video

1. Start at 1 and multiply by 2: 2, 4, 8, 16... How many times have you multiplied to reach 256?

2. Work out $2^3 \cdot 2^5$ writing out all the twos. How many twos do you get? And $2^8 \div 2^3$?

3. Revision of the previous class: if $|x - 3| \leq 1$, between which two numbers is $x - 3$? And x?

While you watch the video

1. What interval do the numbers greater than 1 and less than 2 form? Why does it have round brackets?

2. What does it mean to remove the absolute value "on both sides"? Solve $|x - 1| > 2$ step by step.

3. What did Napier promise in 1614? Why did multiplications bother him?

4. What is the base of a power? And the exponent? Which numbers appear in the row of 2, and why isn't 6 there?

5. **Your prediction.** Multiply 8 by 32 with the two rows. Why does it work? Count the twos.

6. In which book from 1544 were the two rows and the same example already there? Which word appears in it for the first time?

7. **The sentence you have to remember.** Complete: finding the logarithm of a number in a base means finding _____.

8. Write the definition with symbols: $\log_a b = c$ means...

9. Why must the base be positive and different from 1?

10. **The product rule** and **the quotient rule**: write them in words and in symbols. Which rules of powers do they come from?

11. Why isn't the logarithm of a sum the sum of the logarithms? Check it with 10 + 10.

12. Work out $\log_2 32$ saying the sentence: "what number must you raise...?".

13. Can a logarithm have a base other than 2? What does "log" with no base mean? What is log 100? And log 1? Why, in any base, is $\log_a 1 = 0$ and $\log_a a = 1$?

14. Do the numbers between 0 and 1, like the fraction 1/10, have a logarithm? Why, to the left, do you divide below and subtract above?

15. **Your bet.** What number must you raise 10 to in order to get 0.01? And what is $\log_2 (1/8)$?

16. Some people say that log 0 = 0. What is wrong? Is there a logarithm of a negative number?

17. Why, to multiply 3 by 7, do you need the logarithms of all numbers? Between which two whole numbers is $\log_2 3$?

18. **Your prediction.** On the slide rule, with the 1 on top over the 2 below, which number is left below the 4? Which two lengths are added?

19. In $\beta = 10 \cdot \log (I / I_0)$, what is I? And I_0? Where does the 10 in front come from?

20. Why is the faintest sound we can hear at 0 decibels? Which of today's rules says so?

21. With the product rule, explain why a sound ten times more intense has 10 decibels more.

The logarithm, with gaps

Work out $\log_3 9$, filling in the gaps without looking at the video.

1 · The usual question. What number must you raise _____ to in order to get _____? That is, 3^◯ = _____.

2 · The times. Start at 1 and multiply by 3: 1 → _____ → _____. You have multiplied _____ times.

3 · The logarithm. $\log_3 9$ = _____, because 3^__ = 9.

Now, without help:

1. $\log_3 (1/9)$. Hint: to the left, each step divides by 3.

2. $\log_5 1$. Do you need to calculate anything?

Practice

1. $\log_3 81$

2. One sound at 80 dB and another at 50 dB: how many times more intense is the first?

3. Between which two whole numbers is $\log_2 10$? Write it as an interval.

4. **One more, with the rules.** Knowing that log 2 ≈ 0.30, work out in your head log 4, log 8, log 5 (hint: 5 = 10 / 2) and log 20.

5. **One more, with sound.** One sound at 85 dB and another at 95 dB: how many times more intense is the second? And one at 75 dB compared with one at 105 dB?

6. **A challenge.** Why is log 2 almost 0.3? Hint: $2^{10} = 1024$, which is almost 1000. Write log 1024 as a sum of ten logarithms.

Keep thinking

A sound ten times more intense: do you hear it ten times louder? Or does your ear also work, in some way, with logarithms?

GinCol Lab worksheet, made with AI (Claude, by Anthropic) and reviewed by a person · CC BY 4.0 · Source and errata